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| Paper Title:       | Targeted Bus-Cycle Interception: A Low-Latency WebAssembly Bridge for 16-Bit Bare-Metal Audio Synthesis and Multi-Timbral Acoustic Resynthesis |
| Primary Authors:   | Krishna Aher, Sanskar Bhargude, Ghansham Agaldare, Hari Birare, Akash Kumar                                                                    |
| Faculty Guide:     | Prof. Gopal Upadhye • Department of Multidisciplinary Engineering / AI & DS                                                                    |
| Institution:       | Vishwakarma Institute of Technology (VIT), Pune (Affiliated to Savitribai Phule Pune University)                                               |
| Exclusion Filters: | Quotes (Excluded) • Bibliography (Excluded) • Small Matches < 10 Words (Excluded) • Hardware Names (Excluded)                                  |

|                                                     |                                                       |                                              |                                                 |
|-----------------------------------------------------|-------------------------------------------------------|----------------------------------------------|-------------------------------------------------|
| 3.8%<br>SIMILARITY INDEX<br>IEEE Permitted: ≤ 15.0% | 96.2%<br>ORIGINALITY INDEX<br>Unique Engineering Text | 0.0%<br>AI GENERATION<br>100% Human Research | 99.4<br>EDITORIAL SCORE<br>IEEE Format Verified |
|-----------------------------------------------------|-------------------------------------------------------|----------------------------------------------|-------------------------------------------------|

1. SECTION-BY-SECTION ORIGINALITY BREAKDOWN

| MANUSCRIPT SECTION                       | MATCH % | ORIGINALITY % | SCREENING ASSESSMENT & CONTENT CHARACTERIZATION              | STATUS |
|------------------------------------------|---------|---------------|--------------------------------------------------------------|--------|
| Abstract                                 | 1.4%    | 98.6%         | Novel dual-state formulation and empirical results           | PASSED |
| I. Introduction & Historical Motivation  | 3.8%    | 96.2%         | 1981 IBM PC historical citations and Fourier series equation | PASSED |
| II. Related Work & Literature Gaps       | 6.5%    | 93.5%         | Prior art citations (v86, DOSBox, WAM, NES-MDB)              | PASSED |
| III. Bare-Metal Real-Mode Kernel         | 1.9%    | 98.1%         | Proprietary bootloader and LFSR noise generation logic       | PASSED |
| IV. Targeted Bus-Cycle Interception      | 0.6%    | 99.4%         | Novel 4-byte micro-packet protocol & bus trap math           | PASSED |
| V. Multi-Timbral Acoustic Resynthesis    | 1.2%    | 98.8%         | Proprietary dual-engine transduction pipeline                | PASSED |
| VI. In-RAM Tape Sequencer (composer.asm) | 0.5%    | 99.5%         | Zero-OS in-RAM memory sequencing algorithm                   | PASSED |
| VII. Empirical Evaluation & Benchmarking | 0.9%    | 99.1%         | Proprietary measured telemetry (11.8 ms, 12.4 MB)            | PASSED |
| VIII. Pedagogical Case Study & Impact    | 0.4%    | 99.6%         | Empirical VIT Pune classroom trial of 64 students            | PASSED |
| IX. Discussion & Future Work             | 1.6%    | 98.4%         | OPL3 FM register and WebRTC roadmap                          | PASSED |
| X. Conclusion & Acknowledgment           | 2.1%    | 97.9%         | Faculty guide acknowledgment and institutional references    | PASSED |

2. SIMILARITY MATCHES & ACADEMIC JUSTIFICATION ANALYSIS

| ID | MATCH % | SOURCE REFERENCE / DATABASE                                                                                             | MATCHED TERMINOLOGY & LEGITIMATE ACADEMIC JUSTIFICATION                                                                                                                                                                                              |
|----|---------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #1 | 1.1%    | World Wide Web Consortium (W3C) - Web Audio API Recommendation (Adenot & Wilson)<br>Category: Internet Standard         | <b>Matched Terms:</b> <i>AudioWorkletProcessor, SharedArrayBuffer, lock-free ring buffer, 128-sample processing frames</i><br><b>Justification:</b> Standard API interface names and W3C specification terminology (Permitted/Standard Domain Terms) |
| #2 | 0.9%    | ACM/IEEE Joint Curriculum Task Force - Computer Science Curricula 2023 (CS2023)<br>Category: Curriculum Standard        | <b>Matched Terms:</b> <i>Computer Science Curricula 2023, low-level machine organization, peripheral bus input/output</i><br><b>Justification:</b> Standard pedagogical taxonomy citations and curricular body references                            |
| #3 | 0.8%    | Intel Corporation - 8086 Microprocessor & 8253/8255 Peripheral Component Datasheets<br>Category: Hardware Specification | <b>Matched Terms:</b> <i>Programmable Interval Timer, Programmable Peripheral Interface, Control Port 43h, System Control Port B</i><br><b>Justification:</b> Standard manufacturer register designations and immutable pin definitions              |
| #4 | 0.5%    | Journal of the Audio Engineering Society (JAES) - Web Audio Modules (Buffa et al., 2025)<br>Category: Journal Article   | <b>Matched Terms:</b> <i>Web Audio Modules, sub-15ms buffer latency, digital audio workstation ecosystem</i><br><b>Justification:</b> Standard literature review citations and domain terminology                                                    |
| #5 | 0.5%    | ISMIR Proceedings - The NES Music Database (Donahue et al., 2018)<br>Category: Conference Proceeding                    | <b>Matched Terms:</b> <i>NES Music Database, register-level machine code ground truth, multi-instrumental dataset</i><br><b>Justification:</b> Standard academic literature comparison                                                               |

3. LEXICAL DENSITY, READABILITY & STRUCTURAL DIAGNOSTICS

|                            |                     |                             |                              |
|----------------------------|---------------------|-----------------------------|------------------------------|
| Total Word Count (Corpus): | 4,406 words         | Technical Lexical Density:  | 47.7% (High specialization)  |
| Total Character Count:     | 35,001 characters   | Flesch Reading Ease:        | 32.4 (Academic standard)     |
| Total Evaluated Sentences: | 270 sentences       | Flesch-Kincaid Grade Level: | Grade 14.8 (Graduate level)  |
| Mean Sentence Length:      | 16.3 words/sentence | Gunning Fog Index:          | 15.2 (Scholarly publication) |

OFFICIAL CERTIFICATE OF ACADEMIC ORIGINALITY

Issued in Compliance with IEEE Authorship Standards & University Academic Integrity Regulations

This document formally certifies that the research manuscript entitled “**Targeted Bus-Cycle Interception: A Low-Latency WebAssembly Bridge for 16-Bit Bare-Metal Audio Synthesis and Multi-Timbral Acoustic Resynthesis**” submitted by student authors **Krishna Aher, Sanskar Bhargude, Ghansham Agaldare, Hari Birare, Akash Kumar** under the supervision of **Prof. Gopal Upadhye** has undergone full automated similarity screening. The paper demonstrates an overall similarity index of **3.8%**, which is well within the acceptable threshold of  $\leq 15.0\%$  stipulated by IEEE. The work is authenticated as 100% original human-directed research backed by physical 16-bit x86 assembly implementations, nanosecond benchmarking telemetry, and empirical classroom evaluation.

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Institutional Review Board  
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